

Mangnike Nulixiati

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EDUCATION

Vanderbilt University **Ph.D. in Computer Science** **May 2027**
Courses Highlights: Advanced Machine Learning, Deep Learning, Numerical Methods, Advanced Computational Mechanics

Peking University **B.S. in Computer Science** **Jul 2022**
Courses Highlights: Computer Graphics, Operating Systems, Computer Architecture, Advanced Mathematics

RESEARCH AND WORK EXPERIENCE

Research Assistant, Vanderbilt University **Aug 2022 - Present**

- Building a 3D virtual try-on pipeline using 3D Gaussian Splatting for garment reconstruction and rendering, enabling size-aware and photorealistic garment transfer.
- Designed implicit neural representations (INRs) to compress high-resolution 2D/3D simulation fields, significantly reducing storage while preserving fine-scale structure.
- Implemented scalable synthetic 2D/3D data generation pipelines using parallel FEM solvers (FEniCS) to produce large-scale dataset for deep learning models.
- Built physics-informed neural networks (PINNs) that learn corrective dynamics from fully compressible simulations to improve lower-fidelity Boussinesq models.
- Implemented and evaluated diffusion-based generative models for 3D shape synthesis, benchmarking voxel, point-based, and implicit representations with respect to sample quality and controllability.
- Developing physics-grounded video generation methods by fine-tuning vision-language models with geometric and physical priors for spatially and temporally consistent synthesis.

Teaching Assistant, Vanderbilt University **Aug 2022 - Present**

- Principles Operating Systems - Fall 2022
- Numerical Methods - Fall 2023, Fall 2024
- Quantum Computing - Spring 2024, Spring 2025
- Applied Machine Learning - Spring 2026

Software Engineering Intern, Amazon Web Services (AWS) **May 2025 - Aug 2025**

- Built a user-facing analytics console for real-time visualization of deployment and usage data.
- Built a secure, full-stack E2E pipeline across ingestion, APIs, and console UI, surfacing usage patterns and accelerating user troubleshooting.

Research Intern (remote), University of California, Davis **Jul 2021 - Dec 2021**

- Implemented differentiable elasticity simulations using FEM and Position-Based Dynamics, enabling learning-based control and optimization.
- Explored the Material Point Method (MPM) in fracture simulation with different numerical solvers

Research Intern, Institute of Software, Chinese Academy of Sciences **Jan 2021 - Dec 2021**

- Proposed a semi-analytical surface tension model and integrated into a GPU-accelerated simulation pipeline.
- Applied CUDA-based parallel computing techniques to achieve real-time SPH fluid simulation.
- Contributed to the development of PeriDyno, a physics-based simulation engine.

Undergraduate Research Assistant, Peking University **May 2020 - Dec 2020**

- Implemented mesh and topology optimization in hyper-elasticity simulation
- Contributed to the development of a physics-based simulation engine: [PhysIKA](#)

PUBLICATIONS

Nurshat Mangnike, Alex Shao, David Hyde. "A Practical Implicit Monolithic Mixed Finite Element Method and Implementation for Boussinesq and Compressible Flow." *Journal of Computational Physics*, 2026 (under review).

Nurshat Mangnike, David Hyde. "Toward Improving Boussinesq Flow Simulations by Learning with Compressible Flow." 2024 Platform for Advanced Scientific Computing. PASC, 2024.

Nurshat Menglik, H. Yao, Y. Zheng, J. Shi, Y. Qiao, X. He. "Semi-Analytical Surface Tension Model for Free Surface Flows." 2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW). IEEE, 2022.

SKILLS

Programming Languages: C/C++, Python, C#, MATLAB, JavaScript, TypeScript

Tools & Frameworks: CUDA, PyTorch, TensorFlow, JAX, OpenGL, Houdini, COMSOL, FEniCS, Mathematica, Unity

Languages: Uyghur (Native), Mandarin (Fluent), English (Fluent)

AWARDS & HONORS

- **Dean's Graduate Fellowship, *Vanderbilt University*** **Sept 2022**
- **Russell G. Hamilton Scholar, *Vanderbilt University*** **Sept 2022**
- **1st Place FortyAU Award for VR Project, *Vanderbilt University*** **Dec 2022**
The VR project, 'Accessibility Quest', won the 1st place award (worth \$4,000) in the VR project competition. The project focuses on using VR technology to improve urban accessibility for people with disabilities, providing valuable insights for city designers.